Research Proposal Outline

Robust Spatial-Temporal Fusion: Graph-State Space Models (G-SSM)

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1. Introduction

Modern machine learning for complex systems (Climate, Industry, Finance) requires models that can capture both intricate spatial dependencies and long-range temporal dynamics. While Transformers and standard GNNs are the current baseline, they face significant scaling and robustness issues. I propose to investigate a novel fusion of **Graph Neural Networks (GNNs)** and **State Space Models (SSMs)**, creating a **"Graph-State Space Model" (G-SSM)** framework designed for robust, multi-scale spatial-temporal forecasting.

2. Proposed Research Axes

2.1. 1. Graph-Integrated State Space Transitions

Existing SSMs (like Mamba) operate on 1D sequences. I propose to generalize the SSM state transition matrix A to be a function of the graph topology. By integrating graph-convolutional operators into the state space recurrence, we can capture how latent states "flow" across a network (e.g., thermal propagation in a gas turbine or weather patterns across a grid).

2.2. 2. Robustness to Noisy and Incomplete Graphs

A major challenge in GNNs is sensitivity to edge noise or missing data. I intend to explore how the inherent stability properties of linear state space systems can be used to regularize graph-based embeddings. By treating the graph-state evolution as a dynamical system, we can apply control-theoretic stability constraints to ensure robust inference even when the underlying graph is partially corrupted.

2.3. 3. Probabilistic G-SSMs for Uncertainty Quantification

For high-stakes applications, point estimates are insufficient. I propose to develop probabilistic versions of the G-SSM that provide calibrated **Uncertainty Quantification (UQ)**. Building on my work with hybrid modeling at Sonatrach, I will investigate how domain-priors can be used to bound the uncertainty of the graph-state transitions, particularly in the "tails" of the distribution where extreme events occur.

3. Alignment with TUM CIT

This research aligns with the Data Analytics and ML group's focus on robust and theoretically sound graph learning. I aim to validate these G-SSM architectures on large-scale datasets (Sensor networks, Climate grids), contributing to the development of next-generation foundational models for complex spatial-temporal systems.